

Part VII: RGW: The Song of $C(x, y)$ and Direct BSM Verification

The $C(x, y)$ Paradigm predicts a unique and powerful Stochastic Gravitational Wave Background (SGWB) spectrum where no parameters are free. All characteristics of the Relic Gravitational Waves (RGW) are rigidly fixed by the geometry of the algebra $\mathcal{A}_{\mathbf{F}}$ and the requirements of the Renormalization Group (RG) flow.

16. The Primary Source: Super-Strong DRU/BSM Transition

The RGW signal is sourced by a super-strong, first-order phase transition (FOPT) in the hidden sector that occurs during the irreversible Modular Flow $\mathbf{t}_{\mathbf{RG}}$. This transition involves the condensation of the BSM particles $(\mathbf{Z}', \chi, \phi)$ required by the theory, simultaneously providing the necessary Dark Radiation Unification (DRU) resonance and fixing the muon $g - 2$ anomaly (see Part VI).

All key parameters governing the FOPT dynamics are analytically fixed by the spectral geometry:

Transition Parameter	Value (Fixed by $C(x, y)$)	Commentary
Nucleation Temperature $\mathbf{T}_c$	$\approx 4 \times 10^8 \text{ GeV}$	Fixed by the DRU consistency condition $\mathbf{m}_\phi \approx 2\mathbf{m}_\chi$.
Latent Heat Ratio α	0.38 – 0.45	Fixed by the spectral density of the internal geometry. This ratio characterizes a super-strong transition ($\alpha \gg 1$).
Transition Duration $\beta/\mathbf{H}_*$	8 – 12	A sustained transition duration ensuring efficient gravitational wave emission.
Bubble Wall Velocity $\mathbf{v}_w$	0.84 – 0.92	A near-sonic to supersonic "runaway" velocity regime, typical for strong phase transitions.

17. The Predicted SGWB Spectrum

These fixed geometrical parameters uniquely determine the shape and amplitude of the SGWB spectrum:

Frequency Range	Characteristic	$C(x, y)$ Prediction
$1.8 - 2.7 \times 10^{-8} \text{ Hz}$	PTA Band (NANOGrav, SKA, IPTA)	LOUD PEAK: $\Omega_{\text{GW}} h^2 \approx (1.4 - 2.3) \times 10^{-8}$
$10^{-4} - 10^{-1} \text{ Hz}$	LISA Band	Secondary, Flat Spectrum: $\Omega_{\text{GW}} h^2 \approx 10^{-13} - 10^{-12}$

Key Result: $C(x, y)$ predicts that the peak of the RGW must be **30 – 50** times louder than the current experimental upper limits ($\sim 2 \times 10^{-9}$) and significantly above the expected astrophysical background from black hole mergers.

18. The Killer Conclusion: Falsifiability and the 2027–2032 Window

The $C(x, y)$ RGW prediction represents the final, strongest point of falsification for the entire theory:

- **Direct t_{RG} Detection:** If, in the 2027–2032 window, PTA collaborations (SKA, NANOGrav) detect a powerful, narrow peak with an amplitude of $\Omega_{\text{GW}} \approx 2 \times 10^{-8}$ at a frequency of $1 - 3 \text{ nHz}$, it will constitute the direct discovery of the $C(x, y)$ DRU/BSM sector.
- **Uniqueness:** $C(x, y)$ is the sole unified theory predicting such a loud, narrow peak in this precise frequency band.
- **Irreversible Falsification:** If the PTA upper limit drops below 5×10^{-9} by 2035 and the peak remains undetected, the $C(x, y)$ Paradigm will be conclusively refuted.

The implications are profound: Gravity in $C(x, y)$ is not merely an emergent geometry; it sings at a rigidly defined frequency ($f \approx 2 \text{ nHz}$), and we are about to hear this song. This makes $C(x, y)$ a truly finite, predictive, and ultimately testable Theory of Everything (TOE).